Keywords: Transformer theory attention low-rank structure random matrix theory softmax spectral analysis

Spectral Degeneration of Self-Attention

Under Isotropic Score Geometry, Softmax Collapses the Attention Spectrum; Causal Masking Grows Stable Rank as $O(\log L)$

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arXiv:2611.02778 cs.LG / stat.ML NeurIPS 2026 CUHK Shenzhen

An idealized spectral model of self-attention under isotropic score geometry reveals why attention matrices are approximately low-rank: the softmax nonlinearity amplifies dominant modes while collapsing the bulk of the spectrum to zero. For bidirectional attention, the stable rank stays bounded as sequence length grows; for causal attention, it grows logarithmically, implying that decoder-only models are intrinsically less compressible per sequence position than encoders.

AT A GLANCE

Attention matrices are empirically low-rank, but the theoretical reason for this degeneration is unknown.

Understanding why attention collapses informs principled low-rank compression, regularization design, and the choice between encoder and decoder architectures.

Idealized Gaussian random matrix model for the score matrix, combined with free probability theory to track how softmax distorts the eigenspectrum.

Softmax amplifies dominant eigenvalues while collapsing the bulk; stable rank is $O(1)$ for bidirectional and $O(\log L)$ for causal attention.

Analytical derivations validated on pretrained GPT-2 and BERT attention heads and synthetic random attention matrices.

The Big Picture

Empirical studies of trained Transformers consistently report that attention matrices are approximately low-rank: a small number of singular values carry most of the mass, while the remaining singular values are near zero. Methods such as Linformer, Nystromformer, and FlashAttention-low-rank exploit this empirically, but without theoretical justification for when and why it holds.

Is low-rank structure a consequence of training dynamics, or is it intrinsic to the softmax attention computation? This paper answers the second question affirmatively: even without training, the softmax nonlinearity induces spectral degeneration in the attention matrix when the pre-softmax score matrix has isotropic geometry.

Why Existing Theory Falls Short

Empirical studies document low-rank structure but do not explain its origin; they attribute it to gradient dynamics without analysing the effect of the softmax function itself.

Linear attention theory replaces softmax with a kernel approximation and analyzes the resulting linear map, but this fundamentally changes the operation and does not explain the behavior of softmax attention.

Rank-collapse results from overparameterized network theory apply to weight matrices in MLP layers and do not transfer to the compositional query–key product structure of attention.

The Isotropic Score Model

Let $S \in \mathbb{R}^{L \times L}$ be the pre-softmax score matrix with $S_{ij} = q_i^\top k_j / \sqrt{d}$. The isotropic assumption models S as a Wigner-type random matrix: $$S = \frac{1}{\sqrt{L}} W + \lambda \mathbf{u}\mathbf{v}^\top$$ where W has i.i.d. entries with mean 0 and variance 1, λ is the signal strength, and $\mathbf{u},\mathbf{v}$ represent a planted dominant mode. This models a “generic” attention head in which most query–key pairs are unrelated (W term) but a few share a strong structural alignment (λ term).

Main Theoretical Results

Softmax Spectral Transformation. Let $A = \mathrm{softmax}(S)$. Under the isotropic model, in the limit $L \to \infty$ with fixed d, the empirical spectral distribution of A converges to a law concentrated near zero, with an isolated outlier eigenvalue at: $$\lambda_1(A) \approx \frac{e^{\lambda/\sqrt{L}}}{\mathbf{1}^\top e^{S\mathbf{v}} / L}$$ while the bulk of the spectrum satisfies $\lambda_i(A) = O(1/L)$ for $i \geq 2$. The **stable rank** $\rho(A) = \|A\|_F^2 / \|A\|_2^2$ therefore obeys: $$\rho(A) = O(1) \quad \text{(bidirectional attention)}$$

Causal Attention. The causal mask introduces a triangular structure in S: row i can only attend to columns $j \leq i$. The row-normalization of softmax then has a gradient: earlier rows normalize over fewer tokens. The analysis shows that this asymmetry causes the stable rank to grow as: $$\rho(A_{\text{causal}}) = O(\log L)$$

Proof Overview

The derivation proceeds in three steps:

1. Apply the semicircle law from random matrix theory to characterize the eigenspectrum of S.

2. Use free probability calculus to track the distortion of the spectral measure through the entrywise exponential $\exp(S)$.

3. Show that the row-normalization step of softmax is approximated by a rank-1 update to $\exp(S)$ in the high-dimensional limit, and derive its effect on singular values.

For causal attention, the proof additionally shows that the per-row normalization constants form a monotonically increasing sequence in the row index, leading to a non-uniform reweighting of singular vectors that grows the stable rank logarithmically.

What the Guarantee Says

The spectral collapse results are asymptotic ($L \to \infty$, fixed d) and require the isotropic geometry assumption, which may not hold for specialized attention heads in trained models. The paper validates that the predictions remain qualitatively accurate for finite L (up to $L = 2048$) and for trained heads in GPT-2 and BERT, but provides no non-asymptotic concentration bounds. The results apply to the attention matrix A itself, not to the final representations AV, which may have different rank properties depending on the value matrix V.

Empirical Validation

- **Synthetic:** Stable rank under the isotropic model matches analytical predictions at $L \in \{128, 512, 2048\}$.
- **GPT-2 (causal):** Empirical stable rank grows as $1.3 \log L + 0.7$ across heads, consistent with the $O(\log L)$ prediction.
- **BERT (bidirectional):** Empirical stable rank stays in the range $[1.8, 4.2]$ across heads and all L tested, consistent with the $O(1)$ bound.
- **Head specialization:** The planted outlier λ correctly predicts the magnitude of the dominant singular value across heads with varying degrees of specialization.

Implications for Practice

Low-rank compression: The $O(1)$ stable rank for bidirectional attention justifies using rank-r approximations with r independent of L. For causal models, rank should scale mildly with L.

Architecture choice: Encoder models can achieve the same compression ratio regardless of context length. Decoder models become progressively harder to compress at long contexts.

Regularization: Encouraging isotropic score geometry during training via spectral normalization of \(Q\) and \(K\) projections may stabilize the spectral structure and improve compressibility.

Reference

Mengda Li, Jianfeng Yao. **Spectral Degeneration of Self-Attention under Isotropic Score Geometry.** arXiv:2611.02778. NeurIPS 2026. https://arxiv.org/abs/2611.02778